



Weather Analytics Dashboard — Final Project Overview

1. Problem Statement

Build a **terminal-based (CLI) Weather Analytics Dashboard** that fetches real-time weather and 7-day forecast data for 10 cities using the **Open-Meteo API**, stores the results in CSV files, and provides an interactive switch-based menu for **data analysis** using Pandas. All outputs are **console-printed tables**. The application must implement **proper logging** throughout for debugging, info tracking, and error reporting.

2. Learning Objectives

By completing this project, engineers will demonstrate proficiency in:

- Reading/writing CSV files
 - Validating API data using Pydantic models
 - Structuring code with dataclasses, classes, inheritance, and protocols
 - Making concurrent async API calls using asyncio and httpx
 - Data analysis with Pandas
 - Exception handling, decorators, comprehensions, generators, and type hints
 - Python logging for terminal output and debugging
-

3. API Documentation & Reference

3.1 Open-Meteo — Overview

Item	Details
Website	https://open-meteo.com
Pricing	Free for non-commercial use
API Key	✗ Not required
Rate Limit	10,000 requests/day
Response Format	JSON

3.2 Geocoding API

Purpose: Convert a city name into geographic coordinates (latitude & longitude) required by the weather API.

Item	Details
Documentation	https://open-meteo.com/en/docs/geocoding-api
Base URL	https://geocoding-api.open-meteo.com/v1/search
Method	GET
Auth	None

Query Parameters

Parameter	Type	Required	Description
name	string	 Yes	City name to search (e.g., London)
count	int	No	Number of results to return (use 1 for best match)
language	string	No	Language for results (default: en)
format	string	No	Response format (default: json)

Sample Request

```
GET https://geocoding-api.open-meteo.com/v1/search?
name=London&count=1&language=en&format=json
```

Sample Response

```
{
  "results": [
    {
      "id": 2643743,
      "name": "London",
      "latitude": 51.50853,
      "longitude": -0.12574,
      "elevation": 25.0,
      "feature_code": "PPLC",
      "country_code": "GB",
      "country": "United Kingdom",
      "timezone": "Europe/London",
      "population": 7556900
    }
  ]
}
```

What You Need from the Response

```
results[0].name      → City name (confirmed)
results[0].latitude  → Latitude (needed for weather API)
results[0].longitude → Longitude (needed for weather API)
results[0].country   → Country name
results[0].timezone  → Timezone string
```

⚠ Edge Cases to Handle

- **City not found** → results key may be missing or empty list
 - **Multiple results** → Always pick the first result (count=1)
 - **Special characters** → São Paulo — ensure proper URL encoding (httpx handles this automatically)
-

3.3 Weather Forecast API

Purpose: Fetch current weather AND 7-day daily forecast in a **single API call** for a specific latitude/longitude.

Item	Details
Documentation	https://open-meteo.com/en/docs
Base URL	https://api.open-meteo.com/v1/forecast
Method	GET
Auth	None

Query Parameters

Parameter	Type	Required	Description
latitude	float	✓ Yes	Latitude from geocoding response
longitude	float	✓ Yes	Longitude from geocoding response
current	string	No	Comma-separated list of current weather variables
daily	string	No	Comma-separated list of daily forecast variables
timezone	string	No	Timezone for the response (use value from geocoding)
forecast_days	int	No	Number of forecast days (1-16, default: 7)

Current Weather Variables (use in current param)

Variable	Unit	Description
temperature_2m	°C	Current temperature at 2m height
relative_humidity_2m	%	Current relative humidity
apparent_temperature	°C	Feels-like temperature
weather_code	WMO code	Weather condition code
wind_speed_10m	km/h	Wind speed at 10m height
pressure_msl	hPa	Sea-level air pressure

Daily Forecast Variables (use in daily param)

Variable	Unit	Description
temperature_2m_max	°C	Maximum daily temperature
temperature_2m_min	°C	Minimum daily temperature
precipitation_sum	mm	Total daily precipitation
wind_speed_10m_max	km/h	Maximum daily wind speed
weather_code	WMO code	Dominant weather condition

Sample Request

```
GET https://api.open-meteo.com/v1/forecast?
latitude=51.5085&longitude=-
0.1257&current=temperature_2m,relative_humidity_2m,apparent_temperatur
e,weather_code,wind_speed_10m,pressure_msl&daily=temperature_2m_max,te
mperature_2m_min,precipitation_sum,wind_speed_10m_max,weather_code&tim
ezone=Europe/London&forecast_days=7
```

Sample Response

```
{
  "latitude": 51.5,
  "longitude": -0.120000005,
  "timezone": "Europe/London",
  "current_units": {
    "temperature_2m": "°C",
    "relative_humidity_2m": "%",
```

```

    "apparent_temperature": "°C",
    "weather_code": "wmo code",
    "wind_speed_10m": "km/h",
    "pressure_msl": "hPa"
  },
  "current": {
    "time": "2024-07-15T14:00",
    "temperature_2m": 22.3,
    "relative_humidity_2m": 55,
    "apparent_temperature": 21.8,
    "weather_code": 3,
    "wind_speed_10m": 12.5,
    "pressure_msl": 1015.2
  },
  "daily_units": {
    "temperature_2m_max": "°C",
    "temperature_2m_min": "°C",
    "precipitation_sum": "mm",
    "wind_speed_10m_max": "km/h",
    "weather_code": "wmo code"
  },
  "daily": {
    "time": [
      "2024-07-15", "2024-07-16", "2024-07-17",
      "2024-07-18", "2024-07-19", "2024-07-20", "2024-07-21"
    ],
    "temperature_2m_max": [23.1, 25.4, 22.8, 27.0, 24.5, 21.3, 26.1],
    "temperature_2m_min": [14.2, 15.6, 13.9, 16.1, 14.8, 12.5, 15.3],
    "precipitation_sum": [0.0, 1.2, 5.4, 0.0, 0.3, 8.1, 0.0],
    "wind_speed_10m_max": [15.3, 12.1, 18.7, 10.5, 14.2, 22.0, 11.8],
    "weather_code": [3, 61, 80, 1, 2, 65, 0]
  }
}

```

What You Get — Breakdown

ONE API call per city returns:

current	→ Single object with today's weather snapshot
daily	→ Arrays of 7 elements (one per day)
time[0..6]	→ 7 dates
temperature_2m_max[0..6]	→ 7 max temperatures
temperature_2m_min[0..6]	→ 7 min temperatures
precipitation_sum[0..6]	→ 7 precipitation values
wind_speed_10m_max[0..6]	→ 7 wind speed values
weather_code[0..6]	→ 7 weather codes

Total API Calls Per Full Fetch

Round	Calls	Method	Purpose
Round 1 — Geocoding	10 concurrent	<code>asyncio.gather()</code>	City name → lat/lon
Round 2 — Weather	10 concurrent	<code>asyncio.gather()</code>	lat/lon → current + 7-day forecast
Total	20 calls		

3.4 WMO Weather Codes Reference

Store this as a dictionary constant in your project.

Code	Description
0	Clear sky
1	Mainly clear
2	Partly cloudy
3	Overcast
45	Foggy
48	Depositing rime fog
51	Light drizzle
53	Moderate drizzle
55	Dense drizzle
61	Slight rain
63	Moderate rain
65	Heavy rain
71	Slight snow fall
73	Moderate snow fall
75	Heavy snow fall
80	Slight rain showers
81	Moderate rain showers
82	Violent rain showers
95	Thunderstorm
96	Thunderstorm with slight hail
99	Thunderstorm with heavy hail

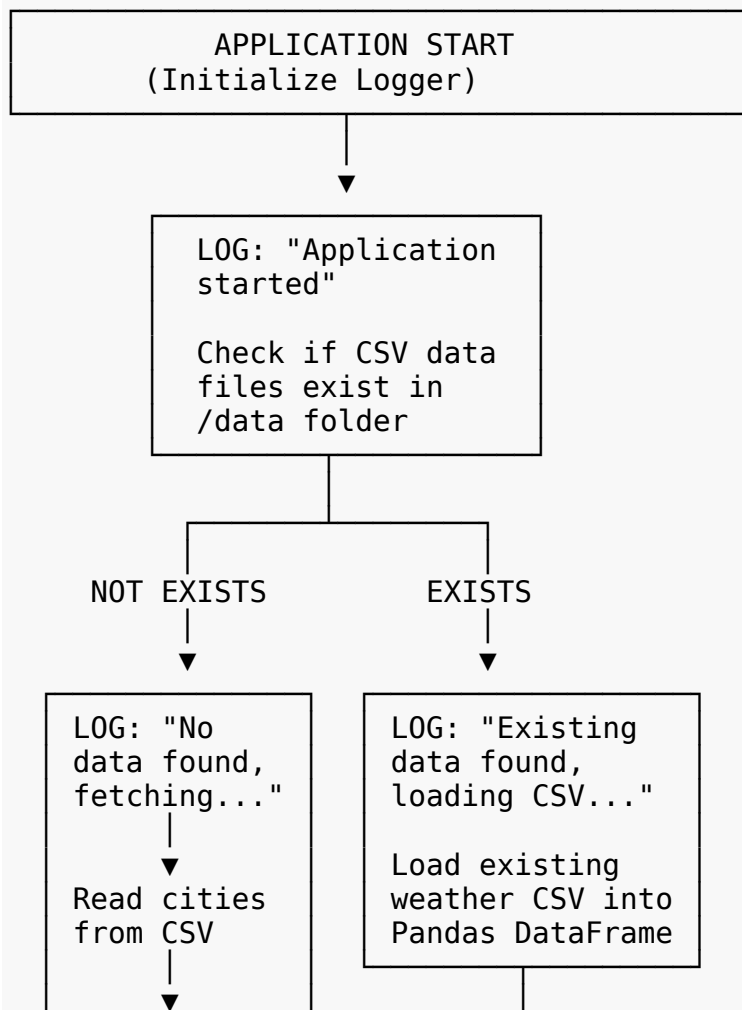
4. Input CSV File

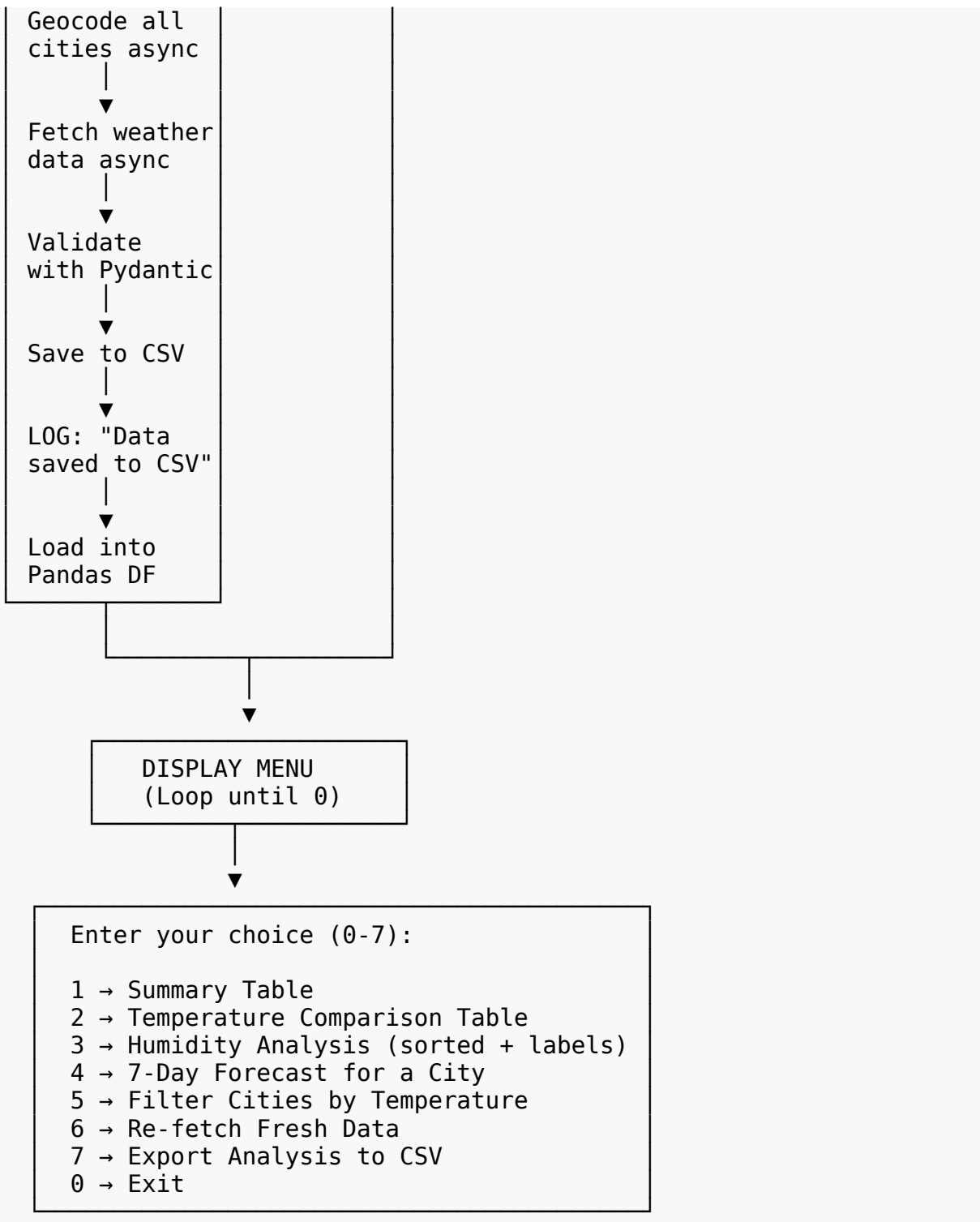
`cities.csv`

```
city,country
London,United Kingdom
New York,United States
Tokyo,Japan
Sydney,Australia
Dubai,UAE
Paris,France
Mumbai,India
Cape Town,South Africa
São Paulo,Brazil
Toronto,Canada
```

Place this file in a /data folder inside the project directory.

5. Application Flow





6. Logging Requirements

Use Python's built-in logging module. All logs should print to **terminal (console)**.

6.1 Logger Setup

Setting	Value
Logger name	weather_dashboard
Handler	StreamHandler (console/terminal output)
Log Level	DEBUG (show all levels)
Format	%(asctime)s %(levelname)-8s %(name)s %(message)s
Date Format	%Y-%m-%d %H:%M:%S

6.2 Sample Log Output

```
2024-07-15 14:30:01 | INFO | weather_dashboard | Application
started
2024-07-15 14:30:01 | INFO | weather_dashboard | Checking for
existing weather data...
2024-07-15 14:30:01 | WARNING | weather_dashboard | No existing data
found. Fetching from API...
2024-07-15 14:30:01 | INFO | weather_dashboard | Reading cities
from data/cities.csv
2024-07-15 14:30:01 | DEBUG | weather_dashboard | Loaded 10 cities
from CSV
2024-07-15 14:30:01 | INFO | weather_dashboard | Starting
geocoding for 10 cities...
2024-07-15 14:30:02 | DEBUG | weather_dashboard | Geocoding London
→ lat=51.5085, lon=-0.1257
2024-07-15 14:30:02 | DEBUG | weather_dashboard | Geocoding New
York → lat=40.7143, lon=-74.006
2024-07-15 14:30:02 | ERROR | weather_dashboard | Geocoding failed
for CityXYZ: City not found
2024-07-15 14:30:02 | INFO | weather_dashboard | Geocoding
complete. 10/10 cities resolved (1.2s)
2024-07-15 14:30:02 | INFO | weather_dashboard | Fetching weather
data for 10 cities...
2024-07-15 14:30:03 | DEBUG | weather_dashboard | Weather data
received for London
2024-07-15 14:30:03 | DEBUG | weather_dashboard | Weather data
received for Tokyo
2024-07-15 14:30:03 | INFO | weather_dashboard | Weather fetch
complete. 10/10 cities (0.8s)
2024-07-15 14:30:03 | INFO | weather_dashboard | Saving current
weather to data/current_weather.csv
2024-07-15 14:30:03 | INFO | weather_dashboard | Saving forecast
```

```

data to data/forecast_weather.csv
2024-07-15 14:30:03 | INFO      | weather_dashboard | Data saved
successfully
2024-07-15 14:30:15 | INFO      | weather_dashboard | User selected
option 1: Summary Table
2024-07-15 14:30:25 | INFO      | weather_dashboard | User selected
option 4: 7-Day Forecast
2024-07-15 14:30:25 | DEBUG     | weather_dashboard | User entered
city: Tokyo
2024-07-15 14:30:35 | WARNING   | weather_dashboard | User entered
invalid city name: XYZ
2024-07-15 14:30:45 | INFO      | weather_dashboard | User selected
option 5: Filter by temperature
2024-07-15 14:30:45 | WARNING   | weather_dashboard | Invalid input for
threshold: 'abc' is not a number
2024-07-15 14:31:00 | INFO      | weather_dashboard | User selected
option 7: Export to CSV
2024-07-15 14:31:00 | INFO      | weather_dashboard | Exported to
data/exports/weather_analysis_20240715_143100.csv
2024-07-15 14:31:10 | INFO      | weather_dashboard | User selected
option 0: Exit
2024-07-15 14:31:10 | INFO      | weather_dashboard | Application
terminated gracefully

```

6.3 When to Use Each Log Level

Level	When to Use	Examples
DEBUG	Detailed internal info useful for development	Individual city geocode results, DataFrame shape, row counts
INFO	General operational events	App started, fetch started/completed, CSV saved, user menu selection
WARNING	Something unexpected but recoverable	Invalid user input, city not found in geocoding, file already exists
ERROR	Something failed that should have worked	API call timeout, network error, JSON parse failure, file write error
CRITICAL	Application cannot continue	Cities CSV not found at all, complete API failure for all cities

7. Switch Operations — Detailed Requirements

Option 1 — Summary Table

Print a formatted console table showing current weather for all 10 cities.

Display Columns:

Column	Source
City	CSV / Geocoding
Country	CSV / Geocoding
Temperature (°C)	current.temperature_2m
Feels Like (°C)	current.apparent_temperature
Humidity (%)	current.relative_humidity_2m
Wind Speed (km/h)	current.wind_speed_10m
Pressure (hPa)	current.pressure_msl
Condition	WMO code → description mapping

Sample Console Output:

City	Country	Temp(°C)	Feels Like	Humidity
Wind(km/h)	Pressure	Condition		
London	United Kingdom	22.3	21.8	55%
12.5	1015.2	Overcast		
Dubai	UAE	41.2	44.5	30%
18.3	1002.1	Clear sky		
...	...	...	...	...
...	...	...		

Logging:

INFO	User selected option 1: Summary Table
DEBUG	Displaying current weather for 10 cities

Option 2 — Temperature Comparison Table

Print a console table comparing actual temperature vs feels-like temperature across all cities, sorted by actual temperature (highest first).

Display Columns:

Column	Description
City	City name
Actual Temp (°C)	temperature_2m
Feels Like (°C)	apparent_temperature
Difference (°C)	apparent_temperature - temperature_2m
Status	"Warmer" if feels like > actual, "Cooler" if less, "Same" if equal

Sample Console Output:

City	Actual(°C)	Feels(°C)	Diff(°C)	Status
Dubai	41.2	44.5	+3.3	Warmer
Mumbai	33.5	38.1	+4.6	Warmer
Toronto	18.7	17.2	-1.5	Cooler
...	...	...	...	...

Logging:

INFO	User selected option 2: Temperature Comparison
DEBUG	Sorted 10 cities by temperature (descending)

Option 3 — Humidity Analysis

Print a console table of humidity levels across all cities, sorted highest to lowest, with text-based severity labels.

Severity Rules:

Humidity Range	Label	Indicator
< 40%	DRY	
40% - 70%	COMFORTABLE	
> 70%	HUMID	

Sample Console Output:

City	Humidity (%)	Severity	Indicator
Mumbai	85	HUMID	
São Paulo	72	HUMID	
London	55	COMFORTABLE	
Dubai	30	DRY	
...	...	...	...

Summary:

-  HUMID cities: 3
-  COMFORTABLE cities: 4
-  DRY cities: 3
-  Average humidity: 54.2%

Logging:

```
INFO | User selected option 3: Humidity Analysis
DEBUG | Categorized 10 cities: 3 HUMID, 4 COMFORTABLE, 3 DRY
```

Option 4 — 7-Day Forecast for a Specific City

Prompt user to enter a city name, then display the 7-day forecast as a console table.

User Interaction:

Available cities: London, New York, Tokyo, Sydney, Dubai, Paris, Mumbai, Cape Town, São Paulo, Toronto

Enter city name: Tokyo

Display Columns:

Column	Source
Date	daily.time[i]
Max Temp (°C)	daily.temperature_2m_max[i]
Min Temp (°C)	daily.temperature_2m_min[i]
Temp Range (°C)	max - min
Precipitation (mm)	daily.precipitation_sum[i]
Max Wind (km/h)	daily.wind_speed_10m_max[i]
Condition	WMO code → description

Sample Console Output:

7-Day Forecast for Tokyo, Japan

Date	Max(°C)	Min(°C)	Range(°C)	Precip(mm)	Wind(km/h)	Condition
2024-07-15	31.2	24.5	6.7	0.0	10.5	Partly cloudy
2024-07-16	33.0	25.1	7.9	2.3	12.1	Slight rain
...	...	...	...	...	...	...
...	...	...	...	...	...	...

Summary:

- 🌡️ Highest: 33.0°C on 2024-07-16
- ❄️ Lowest: 22.1°C on 2024-07-19
- ☁️ Total precipitation: 12.4mm
- 💨 Max wind speed: 22.0 km/h on 2024-07-20

Error Handling: - Invalid city name → Print "City not found. Please enter a valid city name from the list." - Case-insensitive matching (e.g., tokyo should match Tokyo)

Logging:

```
INFO | User selected option 4: 7-Day Forecast
DEBUG | User entered city: Tokyo
INFO | Displaying 7-day forecast for Tokyo
--- OR ---
WARNING | User entered invalid city name: XYZ
```

Option 5 — Filter Cities by Temperature Threshold

Prompt user to enter a temperature value, then display all cities where current temperature exceeds that value.

User Interaction:

Enter temperature threshold (°C): 25

Sample Console Output:

Cities with temperature above 25.0°C:

City	Country	Temp(°C)	Condition
Dubai	UAE	41.2	Clear sky

Mumbai	India	33.5	Partly cloudy
São Paulo	Brazil	28.3	Overcast

3 out of 10 cities match the filter.

Edge Cases: - No cities match → Print "No cities found above {threshold}°C." - Non-numeric input → Print "Invalid input. Please enter a numeric value."

Logging:

```
INFO      | User selected option 5: Filter by Temperature
DEBUG     | User entered threshold: 25.0
INFO      | Found 3 cities above 25.0°C
--- OR ---
WARNING   | Invalid input for threshold: 'abc' is not a number
INFO      | No cities found above 50.0°C
```

Option 6 — Re-fetch Fresh Data

Force re-fetch weather data from API, overwrite existing CSV files, and reload DataFrames.

User Interaction:

△ This will re-fetch all weather data from the API.
Are you sure? (y/n): y

Fetching geocoding data for 10 cities...

✓ Geocoding complete (1.2s)

Fetching weather data for 10 cities...

✓ Weather data fetched (0.8s)

✓ Data saved to CSV files.

✓ DataFrames reloaded successfully.

Timestamp: 2024-07-15 14:32:45

If user enters n:

✗ Re-fetch cancelled. Returning to menu.

Logging:

```
INFO      | User selected option 6: Re-fetch Data
INFO      | User confirmed re-fetch
INFO      | Deleting existing CSV files...
INFO      | Starting geocoding for 10 cities...
INFO      | Geocoding complete. 10/10 cities resolved (1.2s)
```

```
INFO      | Fetching weather data for 10 cities...
INFO      | Weather fetch complete. 10/10 cities (0.8s)
INFO      | Data saved and DataFrames reloaded
--- OR ---
INFO      | User cancelled re-fetch
```

Option 7 — Export Analysis to CSV

Export the current weather summary DataFrame to a new CSV file with a timestamped filename.

Requirements: - Filename format: weather_analysis_YYYYMMDD_HHMMSS.csv - Save in /data/exports/ folder - Create folder if it doesn't exist - Print confirmation with full file path

Sample Console Output:

```
✅ Analysis exported successfully!
📁 File: data/exports/weather_analysis_20240715_143312.csv
📊 Records: 10 cities
```

Logging:

```
INFO      | User selected option 7: Export to CSV
DEBUG     | Creating exports directory: data/exports/
INFO      | Exported 10 records to
data/exports/weather_analysis_20240715_143312.csv
```

Option 0 — Exit

``` 🙌 Thank you for using Weather Analytics Dashboard! Goodbye